[INCH-POUND]
A-A-58039
2 February 1995

COMMERCIAL ITEM DESCRIPTION

VOLTAGE STANDARD

The General Services Administration has authorized
the use of this Commercial Item Description (CID) for all federal agencies

1. SCOPE

1.1 This CID describes a Primary DC Ratio Voltage Standard and is meant as a minimum requirement for the Voltage Standard in which only those manufacturers that meet or surpass the following requirements are qualified per this CID

2 SALIENT CHARACTERISTICS

2 1 Equipment covered by this CID shall be commercially available and may be modified to the extent necessary to meet the description contained herein

2 2 MIL-T-28800E The equipment shall meet the following paragraphs of MIL-T-28800E

Paragraph	Title
3 6.5	Enclosures, Style E
3 7 4 1	Vibration, sinusoidal
3.8 3	EMI, Type III
3.9 3 1	Accessible potentials, Type III
3 9 3 1 2	Leakage current protection, Type III

2.3 Temperature requirements

2.3.1 Non-operating temperature The equipment shall meet the performance and accuracy requirements as specified in ths CID after storage and/or transportation in a non-operating environment of -10°C to 50°C.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to Technology & Industrial Services Division, SA-ALC/TILDD, Bldg. 171, Post C-12, 485 Quentin Roosevelt Rd., Kelly AFB, TX 78241-6425

AMSC N/A FSC 6625

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2.3 2 Operating temperature The equipment shall meet the performance and accuracy requirements as specified in this CID in an operating environment of from 0°C to 23°C

2 4 Mean Time Between Failures (MTBF) The equipment MTBF shall be 2,000 hours The lower test MTBF shall be 2,000 hours and the upper test MTBF shall be 4,000 hours

2 5 Pre-conditioning requirement Each deliverable piece of equipment shall be capable of being subjected to the environmental screening effects of burn-in and thermal cycling, with thermal cycling being accomplished in the sample testing phase

2 6 Calibration and maintenance adjustments The design of the system shall provide for readily accessible calibration and maintenance adjustments These adjustments shall be provided by variable value components which are adjustable by simple means The calibration by substitution of selected components or parts is unacceptable unless specifically approved The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules The calibration interval shall be a period of one year or greater based on an operating time of 2 000 hours

2 7 Performance characteristics The equipment shall be capable of providing ratios between input and output terminals in the following manner

2 7 1 Input selections 1 0 or 1 1

2 7 2 Output settings Between 000 000 0 and 1 099 999 x (@ 1 1 input)

2 8 Ratio standard The equipment shall be able to accommodate voltages of up to 100V while the parasitic thermal electromotive forces (EMFs), which may be present in the standard, shall not exceed 0 5 uv

2 9 Composition The equipment shall consist of the following items and other component parts specified herein and as required to make up a complete equipment

2 9 1 Connector configuration The equipment input and output connectors shall be isolated from the chassis ground No high or low output terminal shall be grounded All input and output connectors shall be of the low thermal EMF type

2 9 2 Ground configuration section The equipment shall be provided with grounding terminals, grounded to the outer chassis

2.9 3 Ratio section. The minimum performance specifications of this section are defined in paragraph 2.9.6, Electrical performance requirements.

2 9 4 Calibration section

2 9 4 1 Calibration bridge The equipment shall be provided with an internal Wheatstone bridge and necessary switches and connections for the calibration of the divider The calibration shall not require any external standard resistor or standard divider, and consist of an equaling

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procedure of the resistors of the first two decades. The procedure may only require a stable 10 V to 20 V DC source and a sensitive null detector, with an input resistance of at least 1 MOhm and capable of resolving of 0.5 uV

2 9 4.2 Resistor adjustments The resistors of the first two decades shall be adjustable through access holes in the front panel The resistors of the third decade shall be adjustable by removing the covers from the instrument The adjustment shall be required not more than once a year Other resistor adjustments shall not be necessary

2 9 4 3 Access to resistor adjustment points The resistor adjustment points shall be covered by a retractable shutter that can be sealed with a calibration sticker to prevent access to unauthorized personnel.

2 9.5 Controls The controls on the equipment shall be rotary knobs The markings on the front panel shall be such that the control function can be easily recognized

2 9 5 1 Bridge balance control The bridge balance control must be concentric with both a fine and coarse range on the same control knob

2 9 6 Electrical performance requirements

2 9 6 1 Ratio range 0 to 1.0 with special terminals for 0 to 1 1

2 9 6 2 Resolution 0 1 Parts Per Million (PPM) of input with seven detented decade dials

2 9 6 3 Absolute linearity +/-0 1 ppm of input at dial settings of 1 1 to 0 1, +/-0 1 x (10S) 1/3 of input at dial settings (S) of 0 1 to 0

2 9 6 4 Absolute linearity stability (Without calibration) +/-1 0 ppm of input/year at dial settings of 1 1 to 0 1, +/- (10S) 2/3 ppm of input/year at dial setting S of 0 1 to 0

2 9 6 5 Temperature coefficient of linearity +/- 0 05 ppm of input/°C maximum at dial settings of 1 1 to 0 1

2 9.6 6 Short term linearity stability With temperature maintained within +/- 1°C, stability of linearity is +/-0.01 ppm/24 hours and +/-0.1 ppm/30 days for applied voltage up to 100V

2.9.6.7 Power coefficient of linearity +/-0.2 ppm of input/W max at dial settings of 1 1 to 0.1, +/-0.2 (10S) 2 ppm of input/W max at settings (S) of 0 1 to 0

2.9.6.8 Maximum end errors. Zero error at output low is +/-0 004 ppm of input Zero error at input low: 0.05 ppm of input. Full scale error. 0.05 ppm of input.

2.9.6.9 Maximum input voltage. 1000 V on 1.0 input terminal, 1100 V on 1 1 input terminal.

2.9.6.10 Maximum carryover error 0 04 ppm

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2 9 6.11 Input resistance 100 kOhms +/-0 005% at 1 0 input terminal at 25°C, 110 KOhms +/-0 005% at 1 1 input terminal at 25°

2 9.6 12 Temperature coefficient of input resistance +/-1 ppm per °C max

2 9 6 13 Thermal EMFs Shall be +/-0.5 uV maximum

2 9 6 14 Breakdown voltage The breakdown voltage shall be a minimum 1500 volts to case to sea level

2 9.7 Cooling devices The equipment shall be capable of operating continuously while drawing maximum power and at the maximum operating temperature, without overheating or causing damage to the equipment Forced air cooling shall not be used

2 9 8 Resistor environment Resistors in a minimum of the first decade A must be immersed in oil to help reduce the effects of self heating and environmental variations The shunt resistor of decade B must also be housed in a mineral oil filled chamber The mineral oil provided as an insulating and cooling medium shall conform to American Society for Testing and Materials (ASTM) Designation D 3487-81 The mineral oil shall conform to the specifications listed for Type I mineral oil listed in this ASTM designation

2 9 9 Self-calibration capability This instrument shall include trimmers, resistors, internal resistance comparator bridges, switching networks and necessary terminals (for connection of external DC source and null detector) to allow precise trimming of the first two decades and checking accuracy of all other decades Trimmers shall be accessible from the front panel

2 10 Additional requirements The manufacturer shall furnish an instruction manual for each instrument It shall contain at least the following operating and maintenance instructions, theory of operation, schematics, pictorial layouts, parts locator diagrams, list of replaceable parts, part numbers, service instruction, etc In addition, all internal assemblies and sub-assemblies shall be permanently labeled in accordance with manual and schematics

3. REGULATORY REQUIREMENTS

3 1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23 403 of the Federal Acquisition Regulation (FAR)

4 QUALITY ASSURANCE PROVISIONS

4.1 Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description,and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices and is the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract

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4 2 Testing The equipment shall be bid sample tested within the accuracies, limits and specifications of this CID It shall be inspected and accepted at the destination as specified in the contract

4 3 The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING

5 1 Preservation, packing, and marking shall be as specified in the contract or order

6. NOTES

6 1 Metric Products Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No 376 and all other requirements of this CID are met If a product is manufactured to metric dimensions exceeding the tolerances specified in the inch-pound units a request should be made to the contracting officer to determine if the product is acceptable

6 2 Federal and Military specifications are available from the Defense Documents Order Desk, Building 4D 700 Robins Avenue Philadelphia PA 19111-5094

6 3 ASTM documents are available from the American Society for Testing and Materials (ASTM) 1916 Race Street, Philadelphia PA 19103

MILITARY INTERESTS

Custodian
Air Force - 99

Preparing Activity
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CIVIL AGENCY COORDINATING ACTIVITY

GSA - FSS

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